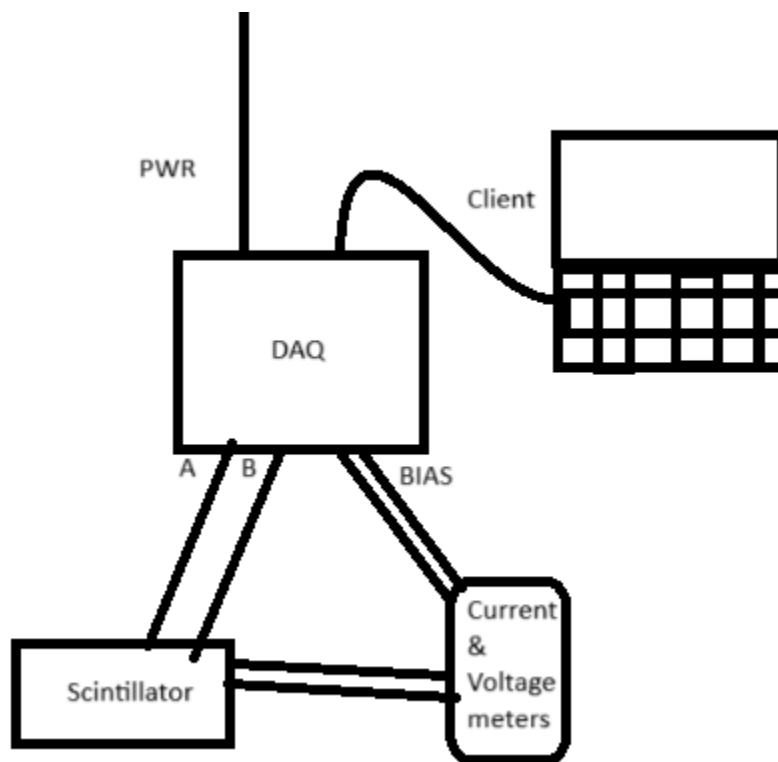


FemtoDAQ LV-2 & Plastic scintillator user manual

Cables:

Connect the DAQ power to your power supply, the DAQ can also use the UBS Client for power (USB client needs to be connected anyways).

Connect the A and B scopes to the plastic scintillator PMT's, now connect the most right pins to the voltage and current meters, the DAQ have a system that limits the current so it shouldn't be an issue, then connect the power from the meters to the scintillator.



Power supply:

Launch the python script and set the wanted voltage, then change the high/low on the device itself and connect to the DAQ, it might also work when reconnection the DAQ from the python script (DO NOT use the reset button on the DAQ, it will require a pc restart to fix it) you can do that with the connect and disconnect buttons in the FemtoDAQ menu in the GUI top left.

Using and setting up the python:

When using the Femto python it might be better to create a copy on your computer storage, it will also help with the .venv.

To set up all of the required Packages you can use pycharm's package installer (pubsub is pypubsub) or install them directly from the freedcamp to the .venv folder (most of the packages in the freedcamp aren't required so it might be faster to select them by hand) you can find them at DAQ -> Femto LV-2 -> .venv -> lib -> site-packages.

The python GUI settings can be changed from the block flow or signal flow, these are the settings we use for now:

Channel A NONINV INVERT Signal Polarity 0 Baseline % Offset	Channel A ☐ Test Enable PULSE Test Select	Channel A ☐ Baseline Restore Enable 10 Baseline Blocking Duration	Channel A RISING FALLING Trigger Edge 500 Trigger Threshold	Channel A ☒ Trigger Enable 10 Trigger Averaging	Channel A NORM Waveform Source	Channel A 1 Histogram Scaling 10 Signal Averaging	Channels A, B 5000 Trigger X position 10 Pileup Detection Duration 10 Pulse Energy Measurement Duration	Channels A, B OR AND Coincidence Logic 10 Global Coincidence Duration	LOW HIGH Voltage Range 60.0 Voltage Setting
Channel B NONINV INVERT Signal Polarity 0 Baseline % Offset	Channel B ☐ Test Enable PULSE Test Select	Channel B ☐ Baseline Restore Enable 10 Baseline Blocking Duration	Channel B RISING FALLING Trigger Edge 500 Trigger Threshold	Channel B ☒ Trigger Enable 10 Trigger Averaging External ☐ Trigger Enable	Channel B NORM Waveform Source	Channel B 1 Histogram Scaling 10 Signal Averaging			

please make sure your computer scale is small enough to see the "update settings" button:

File FemtoDAQ Help

Block Flow Signal Flow Display Waveforms Display Histograms Pulse Energy Logging

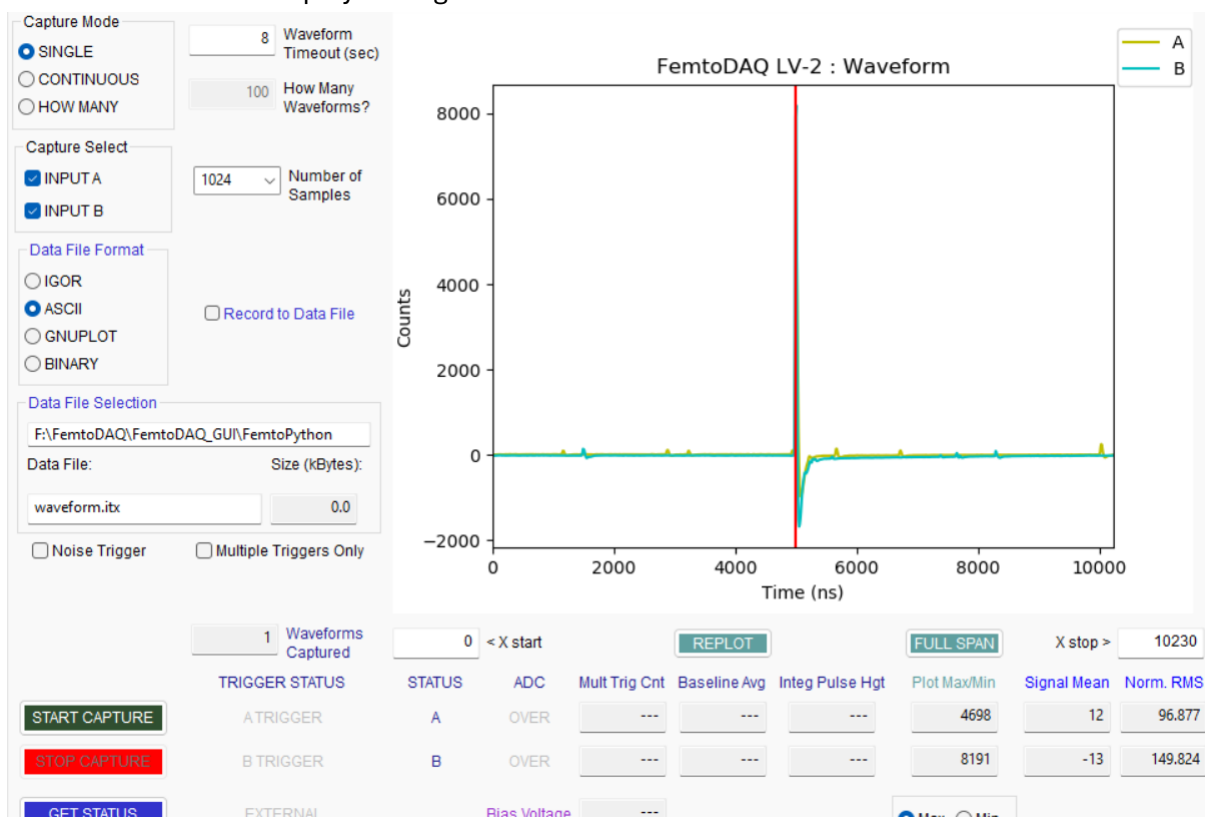
Front End	Internal Test Generator	Baseline Restore	Discriminator	Triggering	Waveform	Histogram	Time Settings	Coincidence	Detector Bias
Channel A IN A → Signal Polarity: NONINV Baseline % Offset: 0	Channel A Test Enable: ☐ Test Select: PULSE	Channel A Baseline Restore Enable: ☐ Baseline Blocking Duration: 10	Channel A Trigger Edge: RISING Trigger Threshold: 500	Channel A Trigger Enable: ☒ Trigger Averaging: 10	Channel A Waveform Source: NORM Histogram Scaling: 1 Signal Averaging: 10	Channel A Trigger X position: 5000 Pileup Detection Duration: 10 Pulse Energy Measurement Duration: 10	Channels A, B Coincidence Logic: OR Global Coincidence Duration: 10	Channels A, B Voltage Range: LOW Voltage Setting: 60.0	
Channel B IN B → Signal Polarity: NONINV Baseline % Offset: 0	Channel B Test Enable: ☐ Test Select: PULSE	Channel B Baseline Restore Enable: ☐ Baseline Blocking Duration: 10	Channel B Trigger Edge: RISING Trigger Threshold: 500	Channel B Trigger Enable: ☒ Trigger Averaging: 10 External Trigger Enable: ☐	Channel B Waveform Source: NORM Histogram Scaling: 1 Signal Averaging: 10				

RETRIEVE SETTINGS UPDATE SETTINGS

Trigger Condition Detected Connected Units: SI

when launching the program please follow the power supply instructions then click FemtoDAQ on the top right and change the Units to SI, then click connect.

here are the current Display settings:



Driver installation:

if you cant connect the DAQ even after restart you might not have the required driver, open the device manager and look for RNDIS, click it and choose update driver, then browse my computer for drivers, let me pick from a list of available drivers on my computer, network adapters, under manufacture choose microsoft, and at model choose Remote NDIS Compatible Device, then next install and restart your computer.